Remote Service & Monitoring using Melsoft GX Works3 & Robustel Gateway Router

Purpose

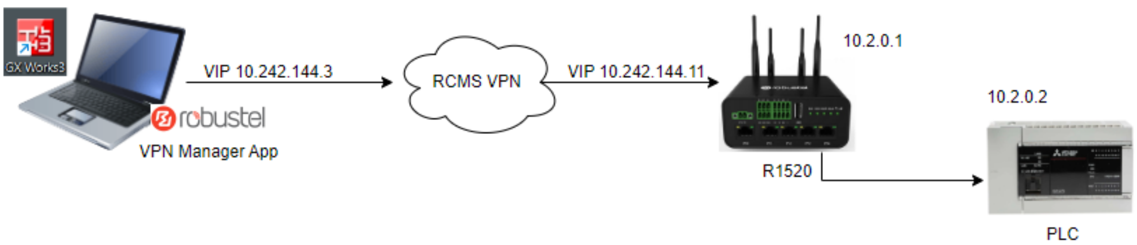
This document provides step-by-step instructions for engineers to remotely access, monitor, and service Mitsubishi PLC's using GX Works3 via a Robustel VPN Router.

It assumes that:

- The Robustel Router has been correctly configured and registered on the RCMS Cloud Platform.
- The router is part of the appropriate VPN Group.
- The network planning and configuration are already completed.
- The Router connect to the Internet and VPN service.

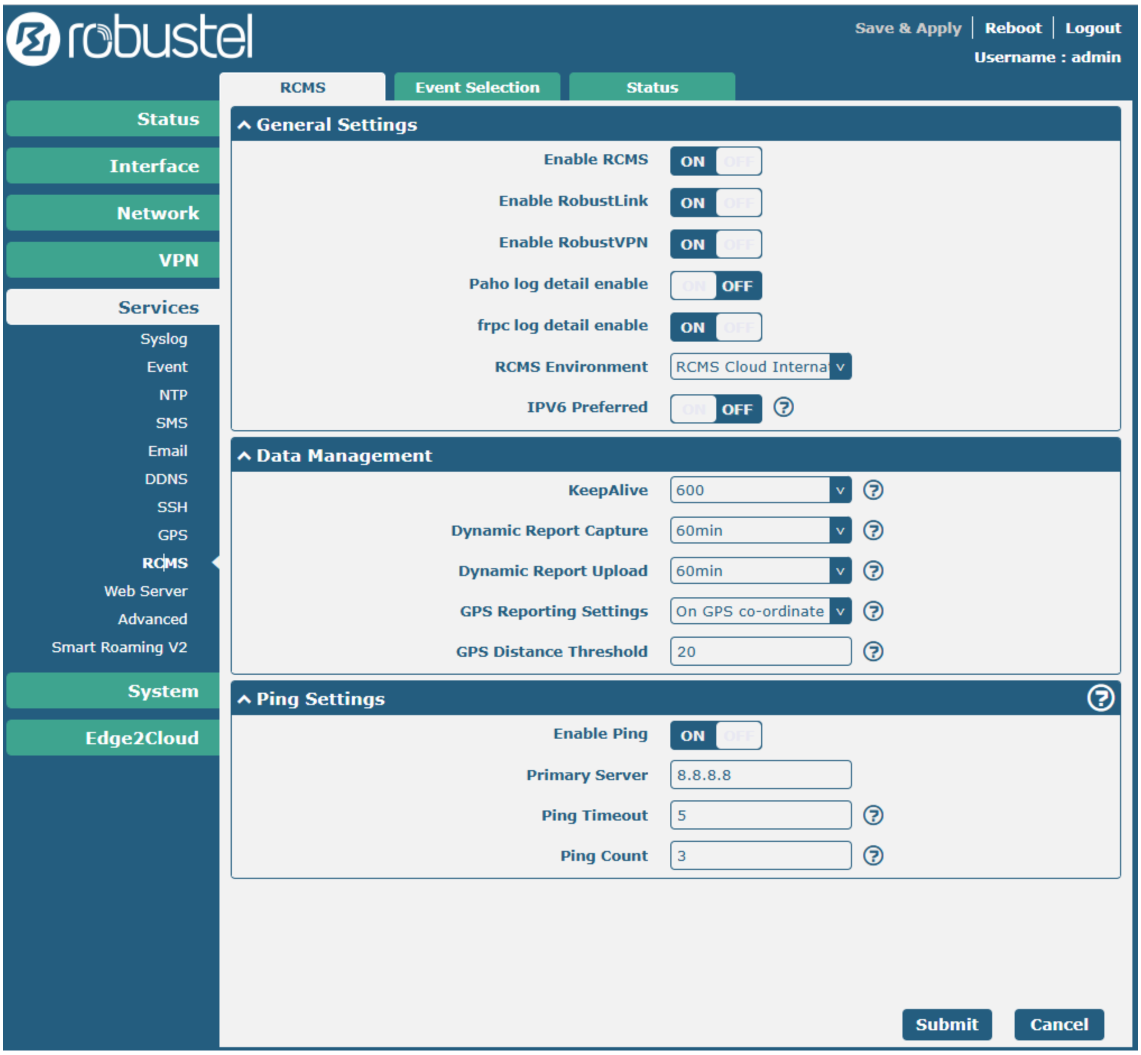
Note: This guide does not cover router configuration or network planning. It focuses on establishing a remote connection to the PLC and using GX Works3 for monitoring or program operations.

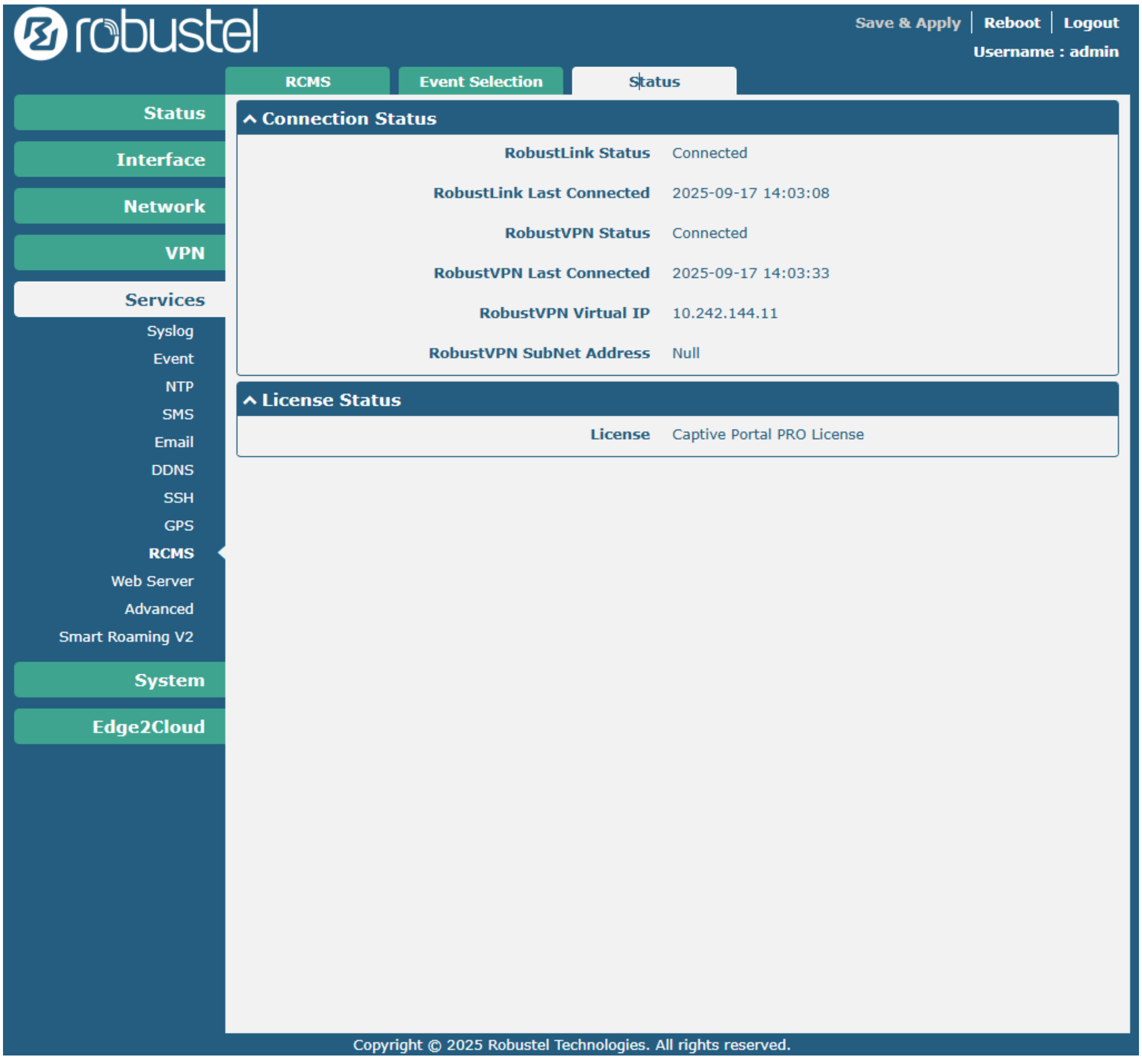
Typical Network Layout



Prerequisites

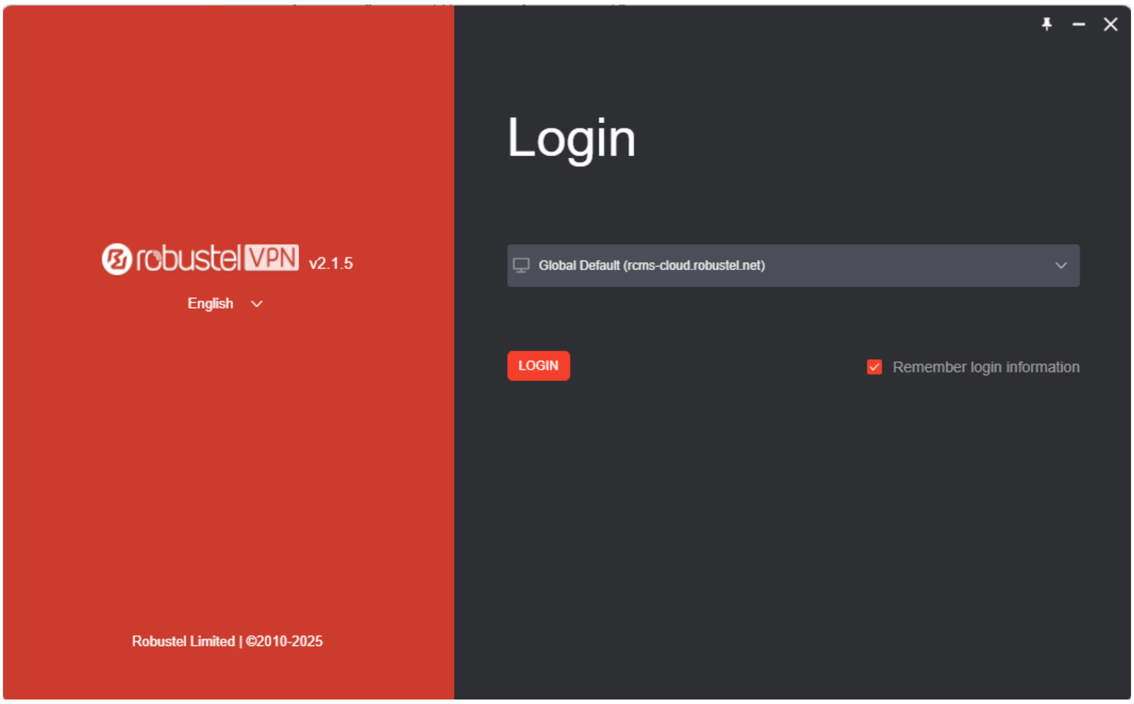
1. The Robustel Router must be registered on the **RCMS Cloud Platform** and included in a VPN group.
2. The **RCMS Client** must be enabled on the router.
3. Verify that the router is connected to the **Robustel VPN** and the link is active.
4. The remote engineer must be a **verified VPN user** in the Robustel RCMS Cloud Platform (https://rcms-cloud.robustel.net/) or request access from the platform administrator.
5. Install the **Robustel VPN Manager** application on your PC. This can be downloaded from the RCMS Cloud Platform (RCMS Support Resources) or requested from the platform administrator.
6. Ensure the PC has a **reliable internet connection**, and your IT administrator allows connections to the Robustel RCMS VPN if using an office network.

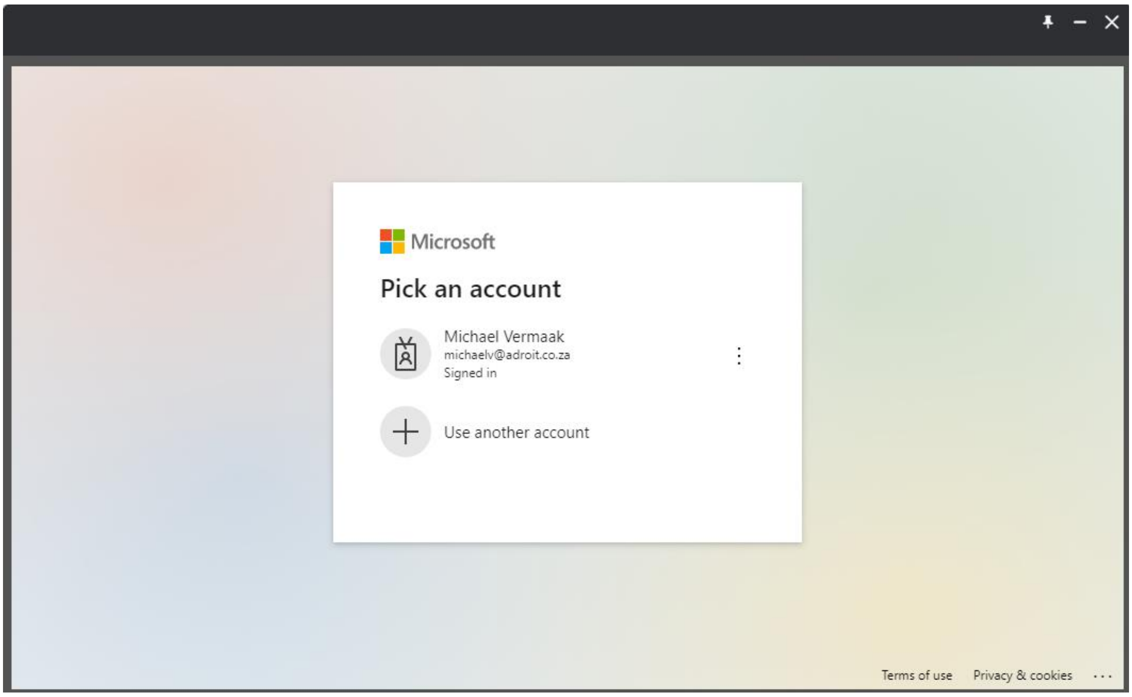

robustel
Save & Apply
Reboot
Logout
Username : admin
RCMS
Event Selection
Status
Status
Interface
Network
VPN
Services
Syslog
Event
NTP
SMS
Email
DDNS
SSH
GPS
RCMS
Web Server
Advanced
Smart Roaming V2
System
Edge2Cloud
General Settings
Enable RCMS
ON OFF
Enable RobustLink
ON OFF
Enable RobustVPN
ON OFF
Paho log detail enable
ON OFF
frpc log detail enable
ON OFF
RCMS Environment
RCMS Cloud Interna
IPV6 Preferred
ON OFF
Data Management
KeepAlive
600
Dynamic Report Capture
60min
Dynamic Report Upload
60min
GPS Reporting Settings
On GPS co-ordinate
GPS Distance Threshold
20
Ping Settings
Enable Ping
ON OFF
Primary Server
8.8.8.8
Ping Timeout
5
Ping Count
3
Submit
Cancel

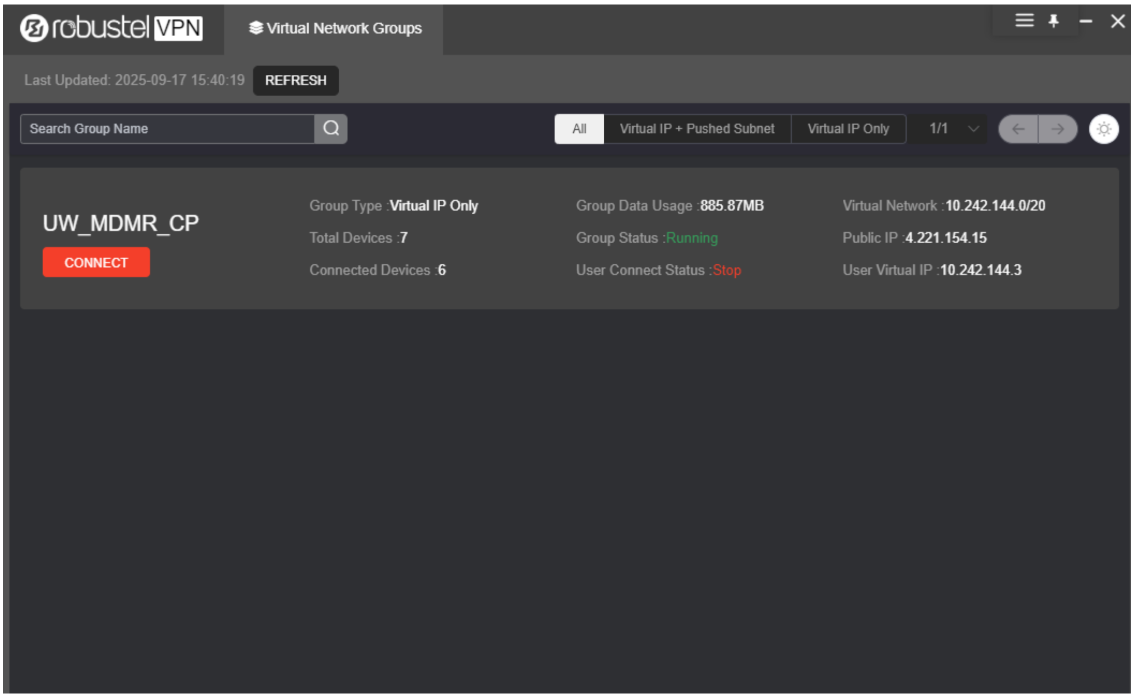


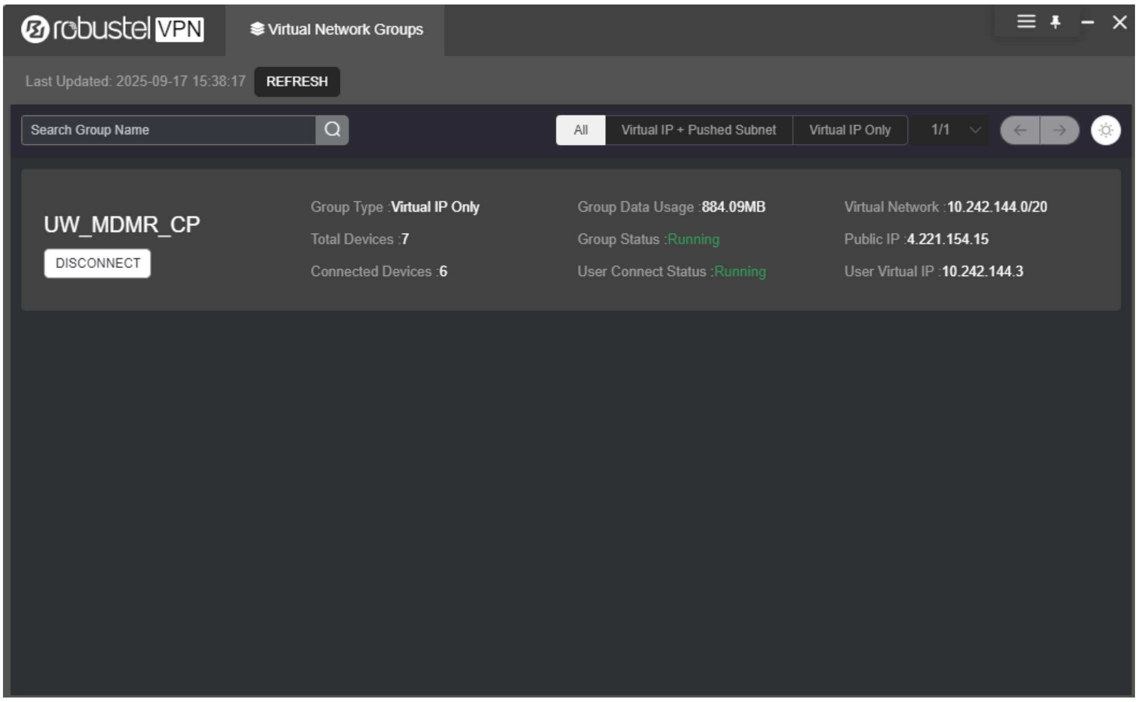
Step 1: Connect to the Robustel VPN

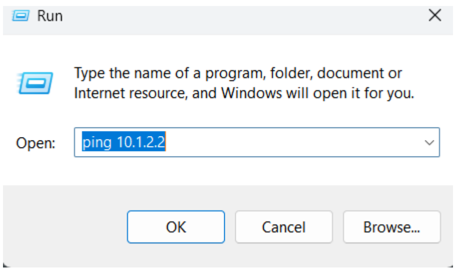
1. Launch the **Robustel VPN Manager** on your PC.
2. Log in using your VPN credentials.
3. If you are part of multiple VPN groups, select the **relevant VPN Group** and click **Connect**.
4. If a **timeout warning** appears, click **Refresh** — the connection may already be established successfully.
5. Ping the PLC to test and ensure connectivity.

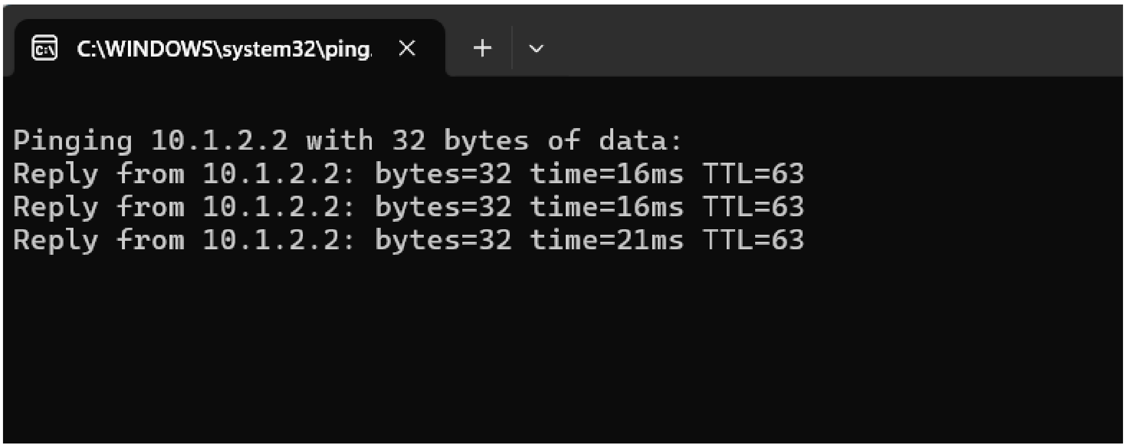

Login
robustel VPN v2.1.5
English
Global Default (rcms-cloud.robustel.net)
LOGIN
Remember login information
Robustel Limited | ©2010-2025


Microsoft
Pick an account
Michael Vermaak
michaelv@adroit.co.za
Signed in
Use another account
Terms of use
Privacy & cookies


robustel VPN
Virtual Network Groups
Last Updated: 2025-09-17 15:40:19
REFRESH
Search Group Name
All
Virtual IP + Pushed Subnet
Virtual IP Only
1/1
UW_MDMR_CP
CONNECT
Group Type :Virtual IP Only
Total Devices :7
Connected Devices :6
Group Data Usage :885.87MB
Group Status :Running
User Connect Status :Stop
Virtual Network :10.242.144.0/20
Public IP :4.221.154.15
User Virtual IP :10.242.144.3

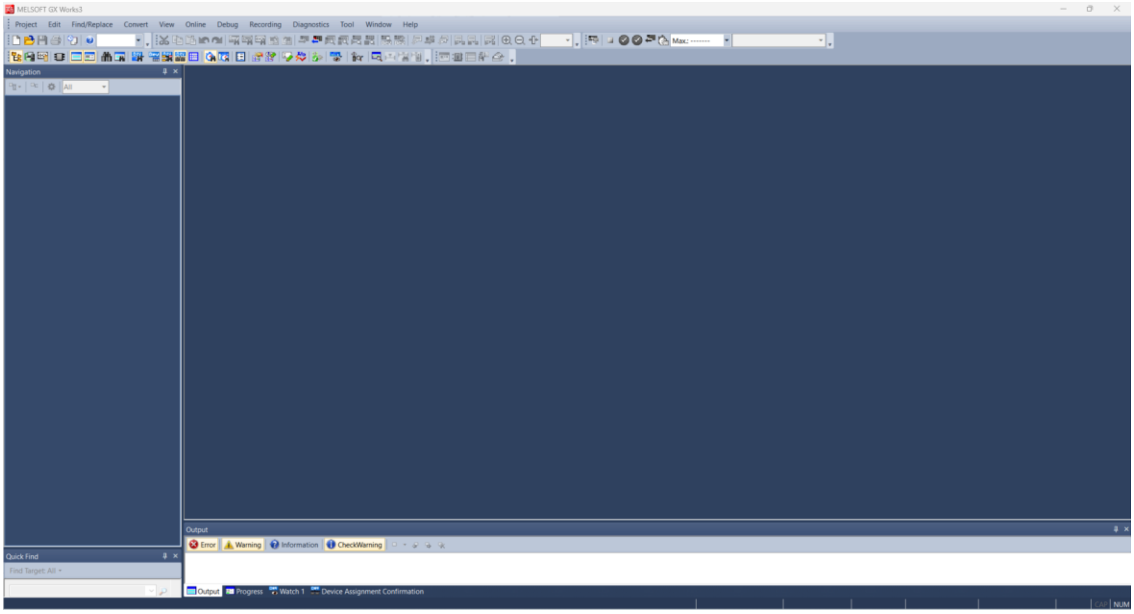

robustel VPN
Virtual Network Groups
Last Updated: 2025-09-17 15:38:17
REFRESH
Search Group Name
All
Virtual IP + Pushed Subnet
Virtual IP Only
1/1
UW_MDMR_CP
DISCONNECT
Group Type :Virtual IP Only
Total Devices :7
Connected Devices :6
Group Data Usage :884.09MB
Group Status :Running
User Connect Status :Running
Virtual Network :10.242.144.0/20
Public IP :4.221.154.15
User Virtual IP :10.242.144.3

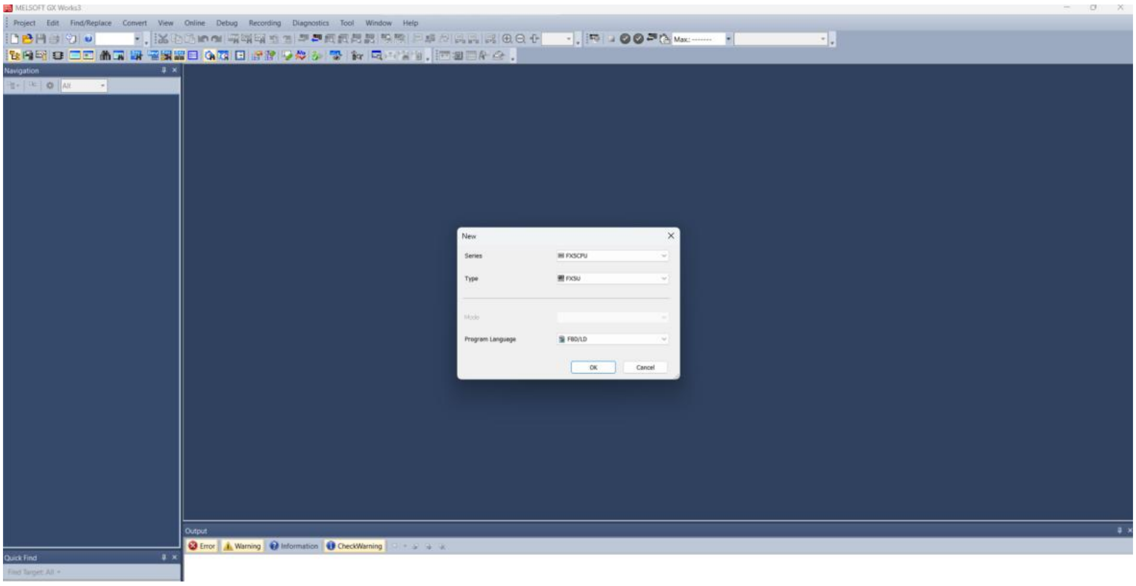

Run
Type the name of a program, folder, document or Internet resource, and Windows will open it for you.
Open:
ping 10.1.2.2
OK
Cancel
Browse...



Step 2: Launch GX Works3

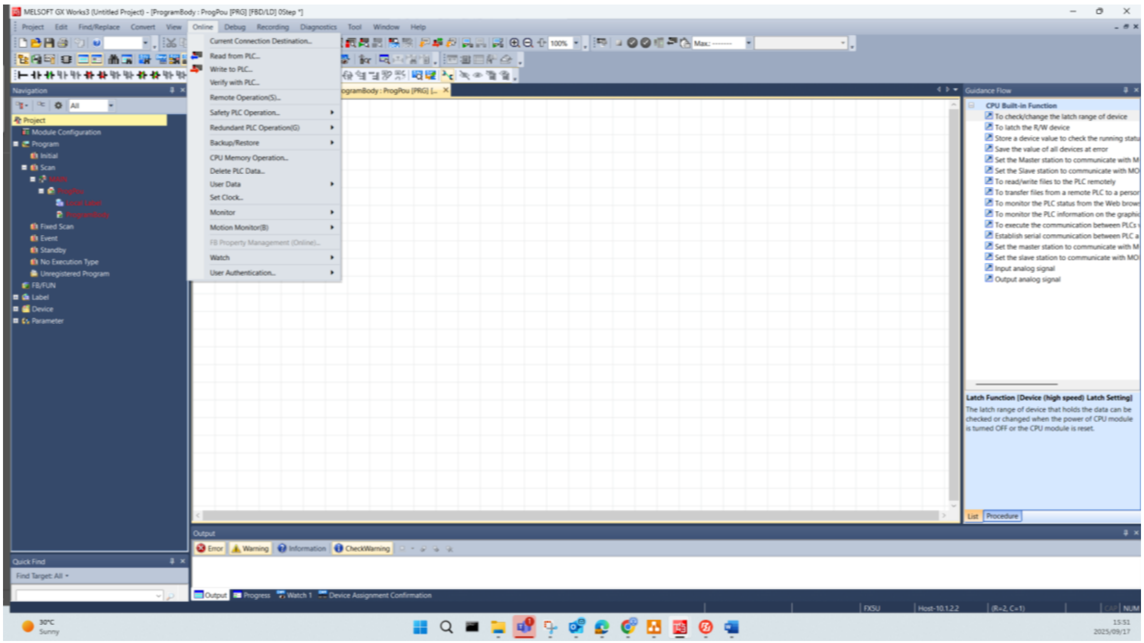
1. Open **GX Works3** on your PC.
2. Select **New Project** or **Load an Existing Project**.



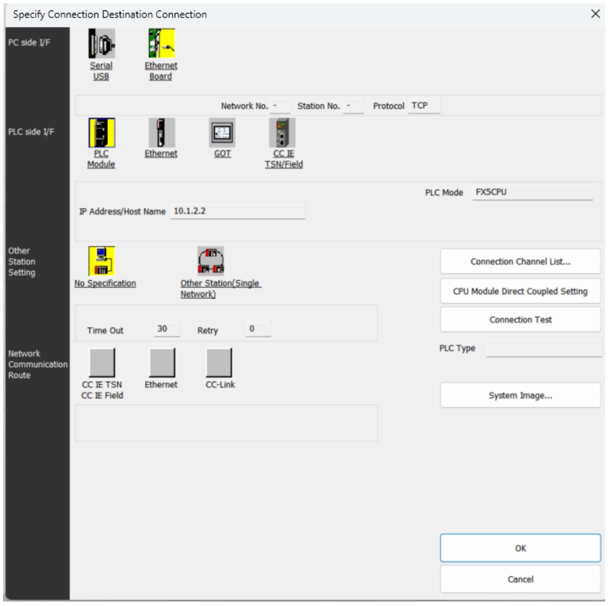


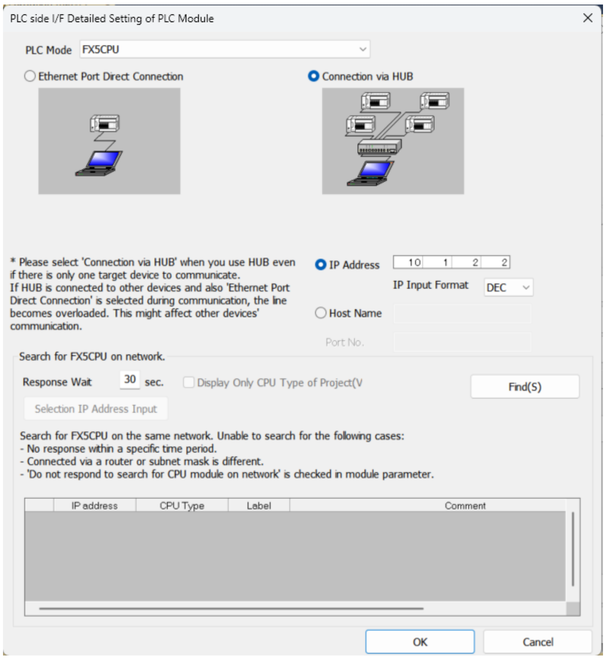
Step 3: Configure PLC Connection

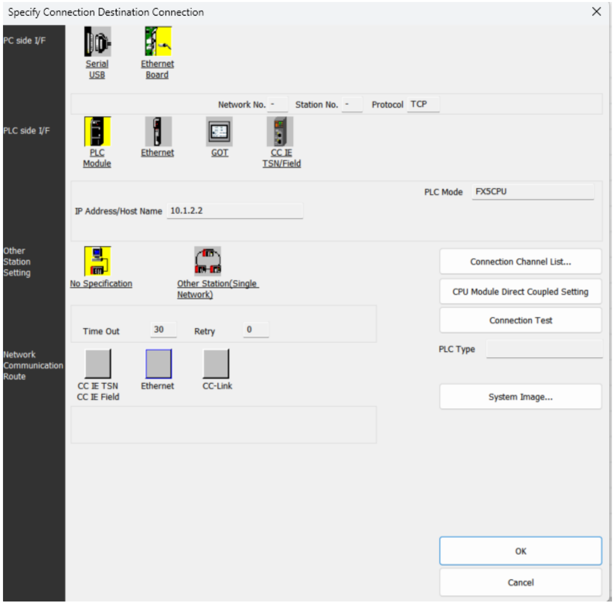
1. From the menu, select **Online → Current Connection Destination**.
2. Choose **Other Connection Method**.
3. Select **Ethernet Board**.
4. Double-click on the **PLC Module**:
    - Configure the **PLC IP Address**.
    - Set the **Response Wait Time**.
    - Click **OK**.
5. Under **Network Communication Route**, select **No Specification** and **Ethernet**.
6. Click **Connection Test**.
    - If successful, the PLC will respond with a message such as: *"Successfully connected with the FX5UCPU"*.
7. Click **OK** to close the connection setup window.

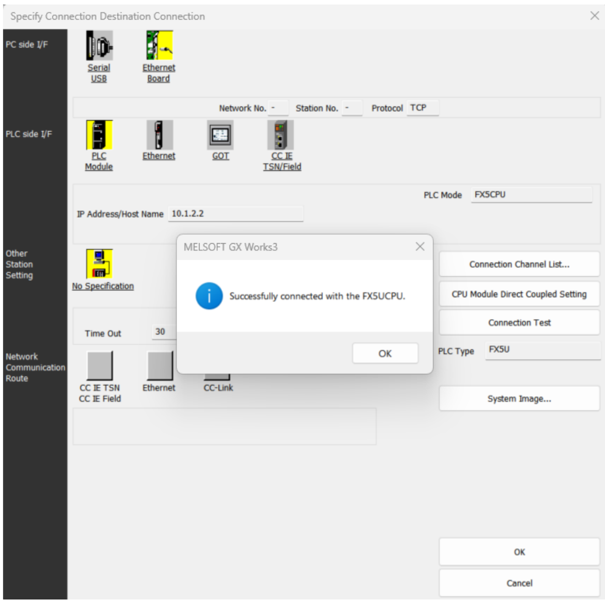



Connection Destination Simple Setting Connection
Direct Coupled Setting
Please select the direct connection method with CPU module.
Ethernet
Ethernet Board
Ethernet
Directly communicate with the CPU module without using a hub.
It is not required to specify the IP address of CPU module.
* This setting is applied to all Ethernet port direct coupled settings.
Adapter
TAP-Windows Adapter V9
IP Address of
10.242.144.3
Communication Test
Other Connection Method
Select this method if you want to connect to CPU module with a connection method other than the direct coupled setting.
Other Connection Method (Open the Specify Connection Destination window)
Do not show this dialog again
* Always open the Specify Connection Destination window.
OK
Cancel


Specify Connection Destination Connection
PC side I/F
Serial USB
Ethernet Board
Network No.
Station No.
Protocol TCP
PLC side I/F
PLC Module
Ethernet
GOT
CC IE TSN/Field
PLC Mode FX5CPU
IP Address/Host Name 10.1.2.2
Other Station Setting
No Specification
Other Station(Single Network)
Connection Channel List...
CPU Module Direct Coupled Setting
Connection Test
Time Out 30
Retry 0
PLC Type
Network Communication Route
CC IE TSN CC IE Field
Ethernet
CC-Link
System Image...
OK
Cancel

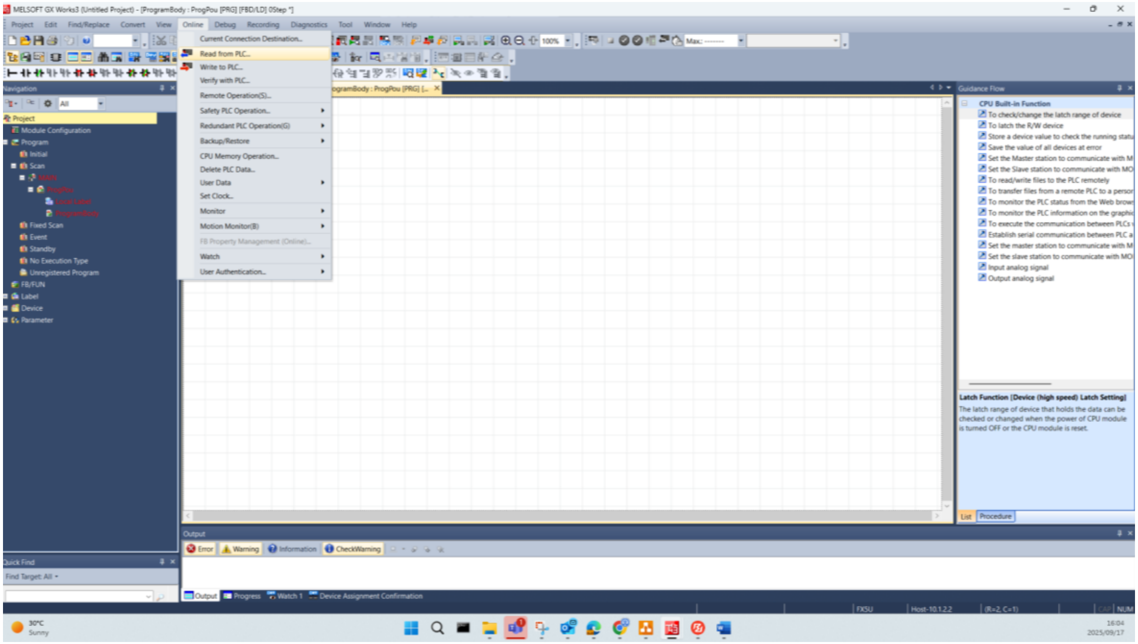

PLC side I/F Detailed Setting of PLC Module
PLC Mode FX5CPU
Ethernet Port Direct Connection
Connection via HUB
* Please select 'Connection via HUB' when you use HUB even if there is only one target device to communicate.
If HUB is connected to other devices and also 'Ethernet Port Direct Connection' is selected during communication, the line becomes overloaded. This might affect other devices' communication.
IP Address 10 1 2 2
IP Input Format DEC
Host Name
Port No.
Search for FX5CPU on network.
Response Wait 30 sec.
Display Only CPU Type of Project(V
Find(S)
Selection IP Address Input
Search for FX5CPU on the same network. Unable to search for the following cases:
- No response within a specific time period.
- Connected via a router or subnet mask is different.
- 'Do not respond to search for CPU module on network' is checked in module parameter.
IP address
CPU Type
Label
Comment
OK
Cancel

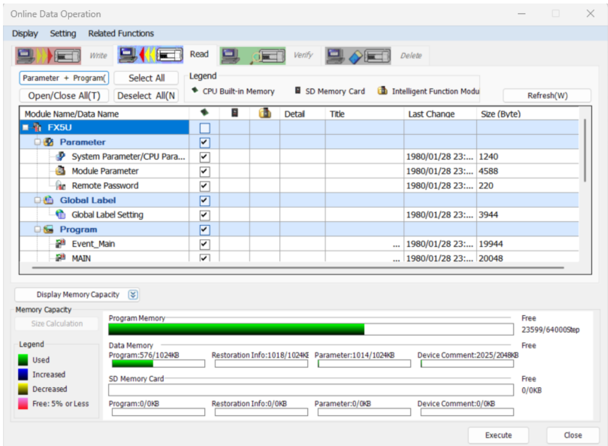


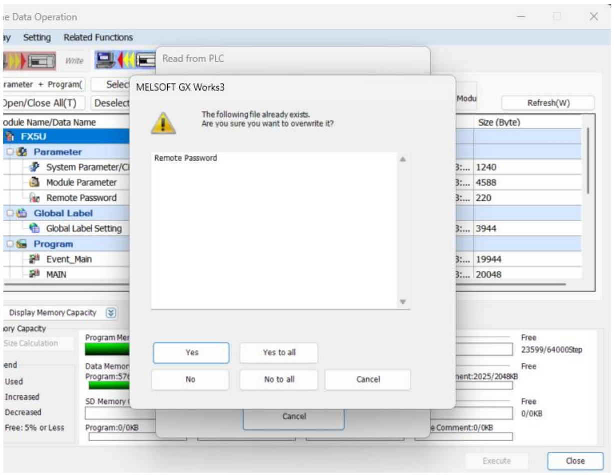


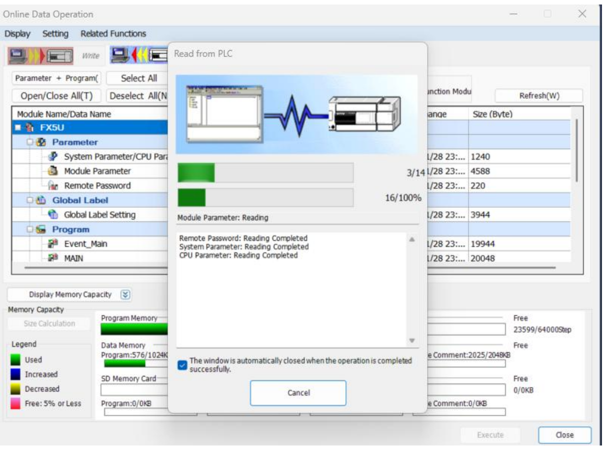
Step 4: Read Parameters from PLC

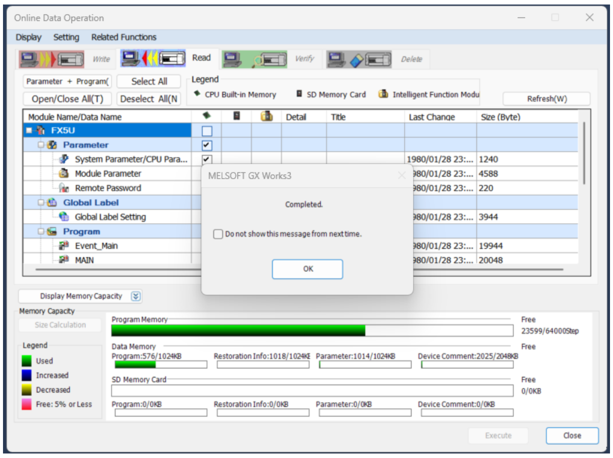
1. Go to **Online → Read from PLC**.
2. Select **Parameters** to read and press **Execute**.
3. Confirm by selecting **Yes to All**.
4. GX Works3 will read the PLC program and parameters.
5. Once complete, press **OK** to close the read window.


Current Connection Destination...
Read from PLC...
Write to PLC...
Verify with PLC...
Remote Operation(S)...
Safety PLC Operation...
Redundant PLC Operation(G)
Backup/Restore
CPU Memory Operation...
Delete PLC Data...
User Data
Set Clock...
Monitor
Motion Monitor(B)
Watch
User Authentication...


Online Data Operation
Display
Setting
Related Functions
Read
Parameter + Program(
Select All
Open/Close All(T)
Deselect All(N
Refresh(W)
CPU Built-in Memory
SD Memory Card
Module Name/Data Name
Detail
Title
Last Change
Size (Byte)
FX5U
Parameter
System Parameter/CPU Para...
Module Parameter
Remote Password
Global Label
Global Label Setting
Program
Event_Main
MAIN
Display Memory Capacity
Memory Capacity
Program Memory
Data Memory
SD Memory Card
Legend
Used
Increased
Decreased
Free: 5% or Less
Execute
Close

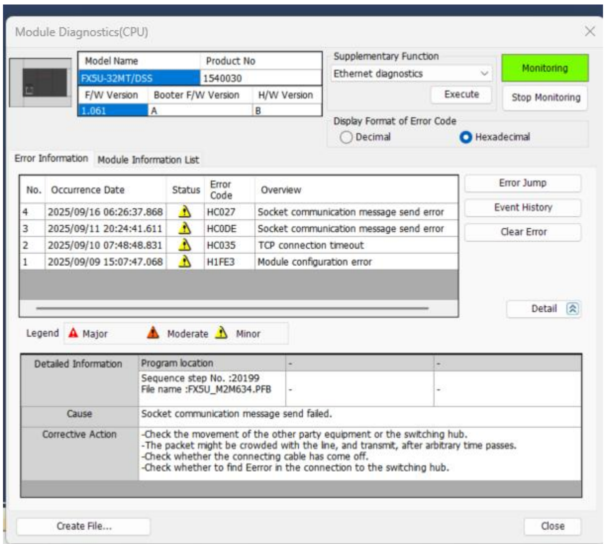

Read from PLC
MELSOFT GX Works3
The following file already exists.
Are you sure you want to overwrite it?
Remote Password
Yes
Yes to all
No
No to all
Cancel





Step 5: Monitor or Modify PLC

- You can now **monitor the PLC** or perform tasks in GX Works3 as required.
- **Important:** Always create **backups of the latest PLC program** before making any changes or uploads.



Additional Notes

- Ensure all operations comply with the site’s **safety and access policies**.
- Maintain documentation of any changes performed during remote service.